

PRODUCT DOCUMENTATION

SignalDraft

An SDR research → outreach drafting tool. Enter a prospect and company; the app runs live public research, confirms the entity with Groq, and drafts a short email for human review. Nothing is auto-sent.

Version: 1.0 · Stack: Next.js 15 · Auth.js · MongoDB Atlas · Groq
Audience: builders, SDRs, and reviewers of this codebase
Generated: 23 August 2026

Contents

- 1. 1. Product overview
- 2. 2. Problem statement
- 3. 3. Architecture & stack
- 4. 4. Research & draft pipeline
- 5. 5. Features by surface
- 6. 6. Edge cases
- 7. 7. Draft confidence
- 8. 8. Local setup
- 9. 9. Vercel deployment
- 0. 10. Gmail drafts
- 1. 11. API map
- 2. 12. Key files
- 3. 13. Operations & troubleshooting

1. Product overview

SignalDraft helps an SDR research a named person at a named company and produce a real first-touch email grounded in a checkable public hook. The output is the email the SDR would send — not a pitch for SignalDraft itself.

Core promise:

- **Live process, not a mock** — Wikipedia, Google News RSS, Hacker News, Groq.
- **Exact person + company** — lookalikes and other-employer news are rejected.
- **Human in the loop** — approve / reject / refine; nothing is auto-sent.
- **Prefer a hold over a wrong email** — if no confirmed hook exists, hold.

Surfaces in the app:

Surface	Route	Purpose
Live run	/	One prospect → research stages → draft → review
Bulk	/bulk	CSV upload, sequential research, review queue
Dashboard	/history	Saved runs, KPIs, edit / approve / Gmail
Login	/login	Google sign-in via Auth.js

2. Problem statement

An SDR enters prospect details (e.g. Jeff Bezos at Amazon) and needs an outreach email that cites a relevant public hook. Research is slow and error-prone when company names collide (Cube Global vs Cube Logic) or when the person appears under another employer (Bezos / Blue Origin while researching Amazon).

SignalDraft automates:

1. Public signal gathering
2. Exact-name ranking and Groq entity confirmation
3. Sentiment / impact analysis
4. A short cold email for human review

3. Architecture & stack

Layer	Choice	Notes
App	Next.js 15 (App Router)	React 19, TypeScript
Auth	Auth.js (NextAuth v5) + Google OAuth	JWT sessions; MongoDB adapter for accounts
Data	MongoDB Atlas	Runs and bulk jobs scoped by userId
LLM	Groq only	Default model openai/gpt-oss-20b
Email drafts	Gmail API (optional)	Incremental gmail.compose scope
Hosting	Vercel	Edge-safe middleware via getToken

Auth layout (important for Vercel)

- `auth.config.ts` — Edge-safe config only (providers: [], authorized callback).
- `auth.ts` — Google provider + MongoDB adapter (Node runtime).
- `middleware.ts` — uses `getToken` from `next-auth/jwt` (avoids middleware crashes).

Data collections

- `Auth.js`: users, accounts, sessions, ...
- `App`: runs, bulk_jobs

Older local `data/runs.json` history is not migrated into MongoDB.

4. Research & draft pipeline

Stage	What happens
Intake	Validate name + company; capture LinkedIn / notes / offer
Company + LinkedIn	Wikipedia snapshot (quoted exact name); optional LinkedIn context
News & funding	Google News RSS (quoted company + soft tokens + offer themes)
Person signals	Person + company queries; HN (full company phrase)
Soft-rank	Score exact / soft / suspect / person; drop junk & stale items
Groq entity match	LLM confirms which candidates are the right org / person
Groq analysis	Sentiment, business impact, outreach angle, risk flags
Draft	Short cold email + Low / Medium / High confidence
Review	Edit, refine, approve/reject, optional Gmail draft — not sent

Matching rules (summary)

- Multi-word companies are never shortened to the first token (no “Cube” for Cube Global).
- A sendable hook must contain the **exact company phrase**.
- Person-only stories about another employer cannot be the primary hook.
- After Groq, a safety net drops “matches” missing the exact company name.

5. Features by surface

Live run (/)

- Intake form (prospect, company, LinkedIn, sender, offer, notes).
- Live stage timeline with per-step durations.
- Signals list with match tier, eligible flag, sensitive flag.
- Groq entity note + analysis cards.
- Editable subject/body, Refine prompt, Approve / Reject, Gmail button.
- Hold and sensitive callouts when applicable.
- Session restore when switching to Dashboard and back.

Bulk (/bulk)

- CSV template download; required columns: fullName (or name), company.
- Optional: title, linkedinUrl, notes.
- Up to 40 rows; processed one at a time.
- Review queue with Approve / Reject / Gmail; Resume if a batch times out on Vercel.

Dashboard (/history)

- KPIs: Runs, Emails stored, Needs review, Approved.
- Loading skeletons (no flash of 0 / 0 / 0) while fetching.
- Search, select a run, edit, store decision, copy, Gmail.
- Shows draft confidence badge and reason.

6. Edge cases

Full detail also lives in docs/EDGE_CASES.md. The pipeline prefers a **hold** over inventing or guessing the wrong company.

6.1 Lookalike company names

Example: Cube Global vs CUBE / Cube Logic / Black Cube.

- Search and match the full quoted company phrase.
- looksLikeWrongCompany flags near-misses.
- Post-LLM safety net drops matches without the exact phrase.
- HN searches the full name, not the first token.

6.2 Person-company split

Example: Jeff Bezos / Amazon vs Blue Origin news.

- First and last name must appear near each other.
- Sendable hook must name the target company.
- If only other-employer stories exist → hold.

6.3 No confirmed public hook

Example: Low-coverage company with no exact-name news.

- Internal hold note (HOLD – do not send), not a fake outreach email.
- Gmail and Refine disabled; Store hold still allowed for tracking.

6.4 Sensitive / negative news

Example: Layoffs, lawsuit, death, bankruptcy, data breach.

- Prefer a non-sensitive exact-company hook when one exists.
- Otherwise: sober tone, risk flags, confidence capped at Medium, UI caution banner.

Residual risks

- Single-token brands (Apple, Delta, Meta) can still collide — prefer legal names.
- Very common names need a LinkedIn URL for safer identity.
- Hooks older than ~18 months are dropped.

7. Draft confidence

Groq rates each drafted email as **Low**, **Medium**, or **High**, with a one-line reason shown in Live, Dashboard, and Bulk.

Rating	Meaning
High	Hook clearly names this person or exact company; facts are in the research pack; offer bridge is natural
Medium	Company match is solid but the hook is thin/older or the offer link is inferred
Low	Weak, stale, or sensitive hook — or no confirmed entity (hold)

Caps: sensitive hooks and auto-rewritten drafts cannot stay High. Holds stay Low. Heuristic fallbacks (if Groq draft fails) use hook strength, not a model self-rating.

Groq entity check reliability

Entity check runs on Groq. The default model `openai/gpt-oss-20b` is a reasoning model; earlier empty replies were caused by a small token budget spent on internal reasoning. Current client uses a larger `max_completion_tokens` budget, low reasoning effort, and fallbacks to other Groq models when the first reply is empty.

8. Local setup

8.1 MongoDB Atlas

1. Create a free cluster at cloud.mongodb.com.
2. Database Access → user + password.
3. Network Access → allow `0.0.0.0/0` (or your IPs) for local + Vercel.
4. Copy the `mongodb+srv://...` URI; URL-encode special characters in the password.

8.2 Google OAuth

1. Google Cloud Console → OAuth client ID (Web application).
2. Origins: `http://localhost:3000`
3. Redirect: `http://localhost:3000/api/auth/callback/google`

8.3 Environment

```
cp .env.example .env.local
openssl rand -base64 32 # → AUTH_SECRET
```


Variable	Value
AUTH_SECRET	Random secret
AUTH_URL	http://localhost:3000 (value only — no AUTH_URL= prefix)
GOOGLE_CLIENT_ID / SECRET	From Google Cloud
MONGODB_URI	Atlas connection string
MONGODB_DB	signaldraft
GROQ_API_KEY	From console.groq.com/keys
GROQ_MODEL	openai/gpt-oss-20b (optional)

8.4 Run

```
npm install
npm run dev
# open http://localhost:3000 → Continue with Google
```

```
npm test      # edge-case + confidence unit tests
npm run build  # production build
```

9. Vercel deployment

1. Push the repo to GitHub; import as a Next.js project on Vercel.
2. Set the same env vars as local, with `AUTH_URL=https://YOUR_PROJECT.vercel.app`.
3. In Google OAuth, add the production origin and callback URL.
4. Deploy; sign in; run a Live draft.

Production URL used in this project: **`https://signaldraft-sand.vercel.app`**. Do not confuse with unrelated Vercel projects on similar domains.

- Hobby function time limits: Live is fine; large Bulk jobs may need **Resume remaining**.
- Atlas must allow Vercel egress (`0.0.0.0/0` is simplest for a personal project).

10. Gmail drafts

Login only requests email/profile. Gmail compose is requested later when the user clicks **Allow Gmail access** / **Add to Gmail drafts**.

1. Enable **Gmail API** in Google Cloud.
2. OAuth consent: External, Testing; add yourself as a test user; add `gmail.compose` scope.
3. Unverified apps show a warning — Advanced → continue (normal for local/test).

Hold notes cannot be saved to Gmail (UI disabled; API rejects). Tokens are persisted in JWT + MongoDB on sign-in so reconnect works under JWT session mode.

11. API map

Route	Role
/api/auth/[...nextauth]	Auth.js handlers (Node runtime)
/api/runs/start	Start live SSE pipeline
/api/runs	List runs for current user
/api/runs/[id]	Get / patch (approve, reject, save edits)
/api/runs/[id]/refine	Rewrite draft with SDR instructions
/api/bulk	Create / list bulk jobs
/api/bulk/[id]	Job detail
/api/bulk/[id]/process	Process / resume queue
/api/gmail/draft	Create Gmail draft
/api/health	Env / health debug

12. Key files

Path	Responsibility
<code>lib/pipeline.ts</code>	Orchestrates stages for a live / bulk run
<code>lib/research.ts</code>	Wikipedia, Google News, HN fetch
<code>lib/relevance.ts</code>	Exact match, ranking, pickHook, soft candidates
<code>lib/edge-cases.ts</code>	Sensitive / hold helpers (client-safe)
<code>lib/llm.ts</code>	Groq chat, entity resolve, analysis, draft, refine
<code>lib/draft.ts</code>	resolveAndAnalyze, writeDraft, hold draft
<code>lib/gmail.ts</code>	Token refresh + Gmail draft create
<code>auth.ts</code> / <code>auth.config.ts</code>	Auth.js split for Edge vs Node
<code>middleware.ts</code>	Route protection via JWT
<code>docs/EDGE_CASES.md</code>	Edge-case writeup
<code>lib/edge-cases.test.ts</code>	Unit tests for the four edge cases + confidence

13. Operations & troubleshooting

Symptom	Likely fix
Groq entity check failed / heuristic only	Confirm <code>GR0Q_API_KEY</code> ; check logs for empty content / model errors; fallbacks should try other Groq models
<code>model_not_found</code>	Set <code>GR0Q_MODEL=openai/gpt-oss-20b</code> (older instant models were retired)
Login 500 / wrong redirect	<code>AUTH_URL</code> must be the bare URL (<code>https://...</code>), not <code>AUTH_URL=https://...</code>
Middleware invocation failed	Keep Edge-safe middleware (<code>getToken</code>), providers only in <code>auth.ts</code>
Gmail access_denied	App in Testing + test user; do not publish unverified restricted scopes
Gmail API disabled	Enable Gmail API in the Google Cloud project
Wrong company in draft	Exact-match + safety net; prefer hold; add LinkedIn / notes and re-run
Dashboard shows empty zeros briefly	Fixed: loading skeletons until <code>/api/runs</code> returns

Demo path

1. Sign in with Google.
2. Use Jeff Bezos / Amazon (or another public company).
3. Ensure GROQ_API_KEY is set.
4. Research & draft email → narrate stages → open Source links.
5. Approve → open Dashboard (your runs only).

SignalDraft documentation • Confidential to the project owners • Not a send tool — every outreach draft requires human review before use in a mailbox.